

A Proposal of Integrated Mach-Zehnder Interferometer

University of British Columbia, Boyan Yu

Abstract—This document describes a design proposal for a Mach-Zehnder interferometer (MZI) on a silicon photonics platform. By designing and fabricating multiple MZIs for the 1550 nm wavelength and slightly varying their design aspects, we can gain a better understanding of how geometrical factors impact the performance of the MZIs and extract the effective group index of the Silicon-Silicon Oxide platform.

Index Terms—laser optics, Mach-Zehnder interferometer, integrated optics, silicon photonics

I. INTRODUCTION

MACH-Zehnder Interferometer (MZI) is among the first few free-space optical transplanted on integrated silicon photonics devices. It has been soon utilized to find its use in optical switching [1], communications [3], and even in computing [2]. In this letter, we will outline designs for a 1550 nm MZI, and we will check how the Free Spectral Range (FSR) changes as the path difference between the two MZI arms changes. Then, we will perform edge analysis for the design instabilities, and compare the measured FSR and group index with the predicted ranges.

II. THEORY

We consider the MZI as a waveguide with a Y branch split into two branches, each with lengths L_1 and L_2 , and then recombined in a Y branch to an output port. The first Y splits the input field E_i into the two branches, $E_1 = E_2 = \frac{E_i}{\sqrt{2}}$, and $E_{1,2}(x, t) = E_{1,2} \exp j(\omega t - \beta x)$. At the second Y that combines the two branches into the output, we have the output field E_o :

$$\begin{aligned} E_o &= E_1(L_1, t) + E_2(L_2, t) \\ &= \frac{E_i}{\sqrt{2}} \exp j\omega t [\exp -j(\beta L_1) + \exp -j(\beta L_2)], \end{aligned} \quad (1)$$

and the output power P_o (in terms of the input power P_i):

$$\begin{aligned} P_o &= \frac{P_i}{4} |\exp -j(\beta L_1) + \exp -j(\beta L_2)|^2 \\ &= \frac{P_i}{2} [1 + \cos \beta(L_2 - L_1)]. \end{aligned} \quad (2)$$

If we define $\Delta L = (L_2 - L_1)$, then we have $P_o = \frac{P_i}{2} [1 + \cos \beta \Delta L]$.

If we vary the wavelength λ , and hence $\beta = \beta(\lambda)$, the output power will also vary. When we see $\beta \Delta L = 2n\pi$ where n is a large natural number, we have maximum outputting power. Therefore, adjacent maxima of P_o are at $\beta(\lambda) \times \Delta L = 2\pi \times n$ and $\beta(\lambda + \delta\lambda) \times \Delta L = 2\pi \times (n+1)$. The difference $d\lambda$ is

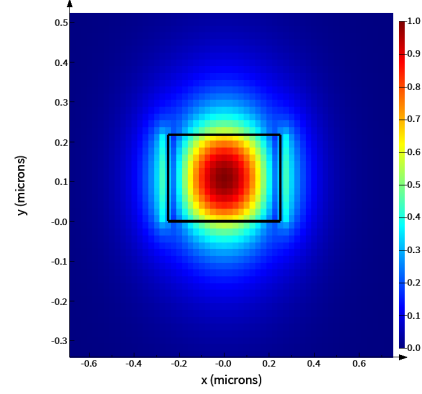


Fig. 1. Simulation of the electrical field intensity in the first TE Mode.

the Free Spectral Range (FSR). Subtracting the two equations, we have:

$$\begin{aligned} \beta(\lambda + \delta\lambda) - \beta(\lambda) &= \delta\lambda \frac{\partial \beta}{\partial \lambda} = \delta\lambda \frac{\partial}{\partial \lambda} \frac{2\pi n}{\lambda} = \frac{2\pi}{\Delta L}; \\ \delta\lambda \left[\frac{1}{\lambda} \frac{\partial n}{\partial \lambda} - \frac{n}{\lambda^2} \right] &= \frac{1}{\Delta L}; \\ \delta\lambda \frac{1}{\lambda^2} \left[\lambda \frac{\partial n}{\partial \lambda} - n \right] &= \delta\lambda \frac{n_g}{\lambda^2} = \frac{1}{\Delta L}. \end{aligned} \quad (3)$$

Therefore, we conclude that for ideal MZIs,

$$\text{FSR} = \delta\lambda = \frac{\lambda^2}{n_g \Delta L}. \quad (4)$$

III. MODELLING AND SIMULATION

We would like to design, model, and simulate MZIs for a wavelength around 1550 nm, in the TE mode.

A. Waveguide

The ridge waveguide we are using is made from Silicon and has a width 500 nm and a height 220 nm. To characterize its performance, we perform a 2D finite difference simulation on a cross-section, where the silicon waveguide is surrounded by Silicon Dioxide (SiO₂). We performed a simulation and obtained the mode profile, as shown in Fig. 1.

We performed a frequency sweep from 1500 nm to 1600 nm with 16 points in between, extracted the effective (n_{eff}) and group index (n_g) at each wavelength point. We perform a curve fit with an order-3 polynomial centered at $\lambda_0 = 1550$ nm, as shown in Fig. 2.

$$n_{\text{eff}} = 2.4468 - 1.1334(\lambda - \lambda_0) - 0.0440(\lambda - \lambda_0)^2 \quad (5)$$

$$n_g = 4.2037 - 0.1355(\lambda - \lambda_0) - 0.1103(\lambda - \lambda_0)^2 \quad (6)$$

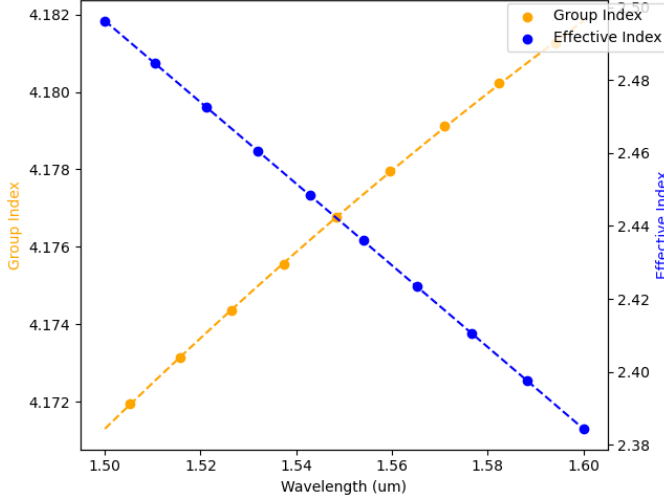


Fig. 2. Effective index n_{eff} simulated and being fitted by a polynomial.

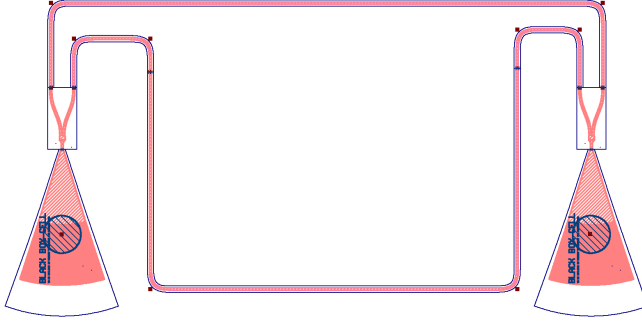


Fig. 3. Layout of a $\Delta L = 90 \mu\text{m}$ MZI, where the length of the left waveguide is 90 μm shorter than the right one. The waveguides, Y branches, and the grating couplers are from the SiEPIC library.

B. Transfer Function

The transfer function of the system is the output-to-input ratio. We have:

$$T(\lambda) = \frac{1}{2} [1 + \cos \beta(L_2 - L_1)]. \quad (7)$$

C. Simulation Results

For the MZIs, we have $\text{FSR} = \frac{\lambda^2}{n_g \Delta L}$. For an MZI with known $\Delta L = L_2 - L_1$, which is varied at design time, we could extract n_g :

$$n_g = \frac{\lambda^2}{\delta \lambda \Delta L} \quad (8)$$

Where $\delta \lambda$, the FSR, is the difference in the transmission minima near the target wavelength, in this proposal, 1550 nm. If we use the calculated n_g values from the previous line fitting, we have $n_g(\lambda_0) \approx 4.2037$, and we could have the following prediction:

We will use the $\Delta L = 90 \mu\text{m}$ case as a demonstration. A similar design layout is shown in Fig. 3. After we have performed a simulation on the individual waveguides, we make use of the standard waveguides, Y branches, and grating couplers given in the SiEPIC library for layout and simulation using ANSYS INTERCONNECT.

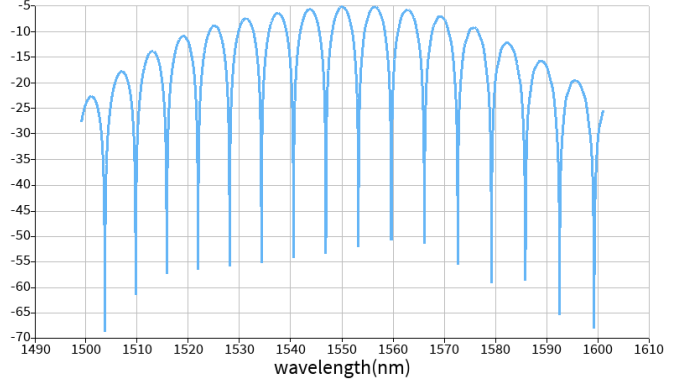


Fig. 4. Simulation of gain from the top coupler (input) to the bottom coupler (output) at different frequencies.

From the plot, we could extract that the FSR is around 18 nm. We could therefore extract the n_g :

$$n_g = \frac{\lambda^2}{\Delta L \delta \lambda} = \frac{1550 \text{ nm} \times 1550 \text{ nm}}{30 \mu\text{m} \times 18 \text{ nm}} \approx 4.449. \quad (9)$$

Or, we could directly compare the FSR with the predicted 19.05 nm.

IV. FABRICATION AND EDGE ANALYSIS

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc., which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness, and 2 μm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 H₂SO₄:H₂O₂) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument at the University of British Columbia. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4" handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2 μm oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300°C. Reflectometry measurements were performed throughout the

Width Height	215.3nm	220nm	223.1nm
470nm	4.22812	4.23253	4.23678
500nm	4.18989	4.19375	4.19777
510nm	4.15797	4.16155	4.16516

TABLE I

SIMULATED GROUP INDICES n_g AT DIFFERENT HEIGHT AND WIDTH COMBINATIONS AT 1550NM WAVELENGTH.

process to verify the device layer, buffer oxide, and cladding thicknesses before delivery.

We understand that the fabrication process introduces inconsistencies with the design. These inconsistencies will mainly manifest as the height and width devices' surfaces using a 10:1 buffered oxide etch, and the devices were inspected with that the wafer thickness varies between 215.3 and 223.1 nm and the width varies between 470 and 510 nm, we could perform simulations on the TM mode at 9 possible combinations of waveguide geometry, and extract the simulated group index n_g . We show the simulated results in Table. I.

And we have seen from Table. I, due to fabrication imperfections, the waveguide group indices range from $4.15797 < n_g < 4.23678$. As previously demonstrated for ideal MZIs, the FSR is inversely proportional to the group index. For each simulated device, we assume that design imperfections will affect the two waveguides of the MZI equally; that is, the two waveguides will always have the same deviations in height and width from the designed values. Therefore, for each of the devices, their respective FSR may be calculated and is tabulated in Table. II.

V. EXPERIMENTAL DATA AND ANALYSIS

To characterize the devices, a custom-built automated test setup [4] [5] with automated control software written in Python was used [6]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [7]. A 90° rotation was used to inject light into the TM grating couplers [7]. A polarization maintaining fibre array was used to couple light in/out of the chip [8].

For each of the six measured devices, we first subtract from a baseline measurement, that corresponds to the losses at each frequency when there is no interference. This is achieved by two reference designs, one with a single waveguide between two grating couplers, and the other one being a balanced MZI. Although there is little difference between them, we will use the balanced MZI as the baseline. Then, we will fit the transfer function

$$10 \lg \left(\frac{1}{4} \left| 1 + \exp \left[-i \frac{2\pi}{\lambda} n_{\text{eff}} \Delta L - \alpha \Delta L / 2 \right] \right|^2 \right) \quad (10)$$

And we have an effective n_{eff} that is also frequency dependent:

$$n_{\text{eff}} = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2 \quad (11)$$

We have shown our measurement and fitting for the device with $\Delta L = 90 \mu\text{m}$ in Figure. 5. As demonstrated, a first order linear equation for n_{eff} already describes the data very well; and we show that the second order dispersion effect is negligible in our analysis. This is true for all MZI devices we have designed, fabricated, and measured. We tabulate our results of the measured group indices and the measured FSRs in the Table II. Since there is little trend in the FSRs and group indices, we will average all our measured values instead of performing a curve fit and extract the value at 1550nm or 1530nm.

To predict the FSR, we apply the equation 8. We consider $\lambda_0 = 1530\text{nm}$, and we use the fitted n_g at this value, because our data is centered on this value; while the standard 1550nm is slightly more noisy, as shown in Figure. 5. To account for the FSR difference due to the manufacture, for each of the FSRs we give a range. We could conclude that all of our data fits in the range of the prediction (FSR and n_g) within their manufacturing variances.

VI. CONCLUSIONS AND OUTLOOK

In this letter, we designed, fabricated, measured, and analyzed integrated MZIs. We have demonstrated that the designing and fabrication is highly reliable, as the manufactured values lie within the expected region in the edge analysis. We believe the effective manufacturing and calibration of integrated MZIs could serve as a basis for future developments for optics in the visible frequency range (for example, 671nm and 780nm, the Lithium and Rubidium resonances). We expect more integrated optical devices will greatly enhance the stability and compactness of visible laser systems in laboratories, and also make them more prone to noises and temperature fluctuations.

REFERENCES

- [1] X. Chen, J. Lin, K. Wang, A Review of Silicon-Based Integrated Optical Switches. *Laser Photonics Rev* 2023, 17, 2200571. <https://doi.org/10.1002/lpor.202200571>
- [2] Marchisio, A., Da Ros, F., Curri, V. et al. Comprehensive model of MZI-based circuits for photonic computing applications. *Commun Phys* 8, 277 (2025). <https://doi.org/10.1038/s42005-025-02176-0>
- [3] Zhou, X., Yi, D., Chan, D.W.U. et al. Silicon photonics for high-speed communications and photonic signal processing. *npj Nanophoton.* 1, 27 (2024). <https://doi.org/10.1038/s44310-024-00024-7>
- [4] Lukas Chrostowski, Michael Hochberg, chapter 12 in "Silicon Photonics Design: From Devices to Systems", Cambridge University Press, 2015
- [5] <http://mapleleafphotonics.com>, Maple Leaf Photonics, Seattle WA, USA.
- [6] <http://siepic.ubc.ca/probestation>, using Python code developed by Michael Caverley.
- [7] Yun Wang, Xu Wang, Jonas Flueckiger, Han Yun, Wei Shi, Richard Bojko, Nicolas A. F. Jaeger, Lukas Chrostowski, "Focusing sub-wavelength grating couplers with low back reflections for rapid prototyping of silicon photonic circuits", *Optics Express* Vol. 22, Issue 17, pp. 20652-20662 (2014) doi: 10.1364/OE.22.020652
- [8] www.plcconnections.com, PLC Connections, Columbus OH, USA.

ΔL (μm)	FSR Range(nm)	Modeled FSR(nm)	Measured FSR(nm)	Measured n_g
40	13.81 , 14.07	14.02	13.748 ± 0.370	4.230 ± 0.047
60	9.21, 9.38	9.35	9.287 ± 0.161	4.175 ± 0.077
90	6.14, 6.26	6.23	6.177 ± 0.125	4.189 ± 0.034
120	4.60, 4.69	4.67	4.593 ± 0.146	4.225 ± 0.150
150	3.68 , 3.75	3.73	3.693 ± 0.107	4.200 ± 0.044

TABLE II

COMPARISON OF MEASURED VALUES OF FSR AND n_g WITH THE THEORETICALLY PREDICTED ONES. WE SHOW THAT OUR FSR AGREES WITH THE MODELS, AND LIE WITHIN THE REGION OF THE FABRICATION IMPERFECTION-INDUCED VARIATIONS IN FSR. WE ALSO SEE THAT OUR MEASURED GROUP INDICES ALL LIE WITHIN THE EXPECTED RANGE IN OUR PREVIOUS EDGE ANALYSIS.

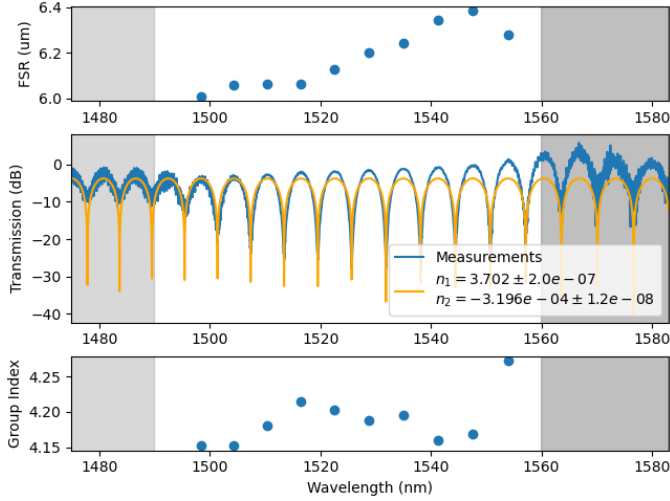


Fig. 5. The Measurements of the $\Delta L = 90\mu\text{m}$ device. Because data in the two edges are noisy, we only fit for the middle unshaded area and extrapolate our fittings to the outside. Middle: the extracted measurement of the transmission in dB (blue) and the fitted function (yellow) according to equation 10. We have fitted n_1 and n_2 parameters, as shown in 11. Top: The extracted FSR from the difference in the dip positions in the measurement data. Bottom: the extracted group index n_g from the FSR according to 8. The above measurements shows that there is little systematic trends in n_g , and hence we consider the second order dispersion to have little effect, and it reflects in $n_3 = 0$ in our fitting.